AI/ML Minor Project Report

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Project Title: End-to-End Image Classification and Web Interface

Dataset Used: MNIST (Handwritten Digits)

Streamlit App Topic: Student Marks Calculator

1. Objective

The objective of this project is to apply deep learning principles by developing a Convolutional Neural Network (CNN) capable of classifying image data, and to demonstrate practical Python development skills by building an interactive data-processing web application. The CNN focuses on computer vision pattern recognition, while the Streamlit application focuses on user input handling, backend logic, and user interface design.

2. CNN Workflow (Part A)

Dataset & Preprocessing: The MNIST dataset was selected for this project. It consists of grayscale images (28x28 pixels) of handwritten digits ranging from 0 to 9. The dataset was loaded using TensorFlow/Keras. The image arrays were reshaped to include a single color channel and the pixel values were normalized (divided by 255.0) to scale them between 0 and 1, ensuring faster and more stable model training.

CNN Approach & Training: A Sequential CNN model was constructed using Conv2D layers for feature extraction, MaxPooling2D layers for downsampling, and Dense layers for final classification utilizing a softmax activation function. The model was compiled using the Adam optimizer and sparse categorical crossentropy loss. The model was trained over 3 epochs.

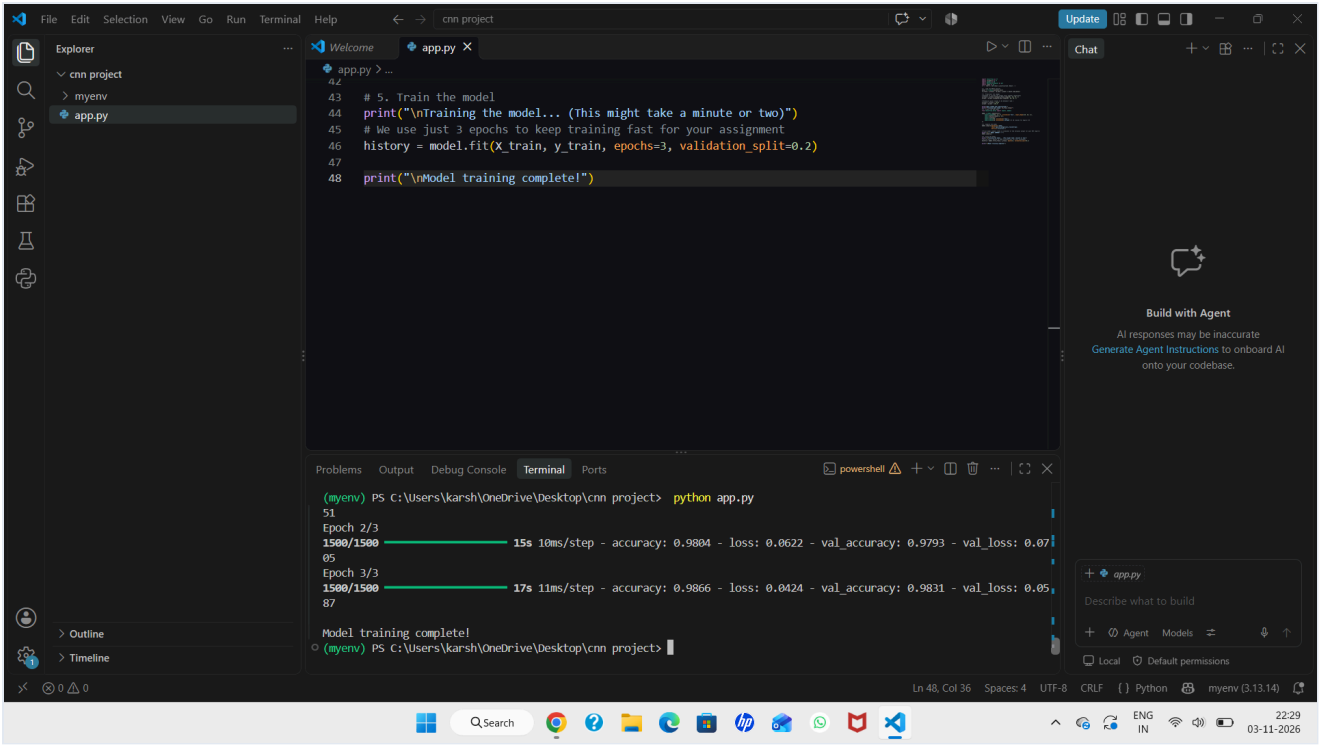


Figure 1: Console output showing successful model training across 3 epochs, achieving over 98% accuracy.

Evaluation & Visualizations: The model's performance was tracked by monitoring both training and validation metrics to ensure it was learning features without overfitting. Sample predictions were subsequently generated on the unseen test data to verify real-world accuracy visually.

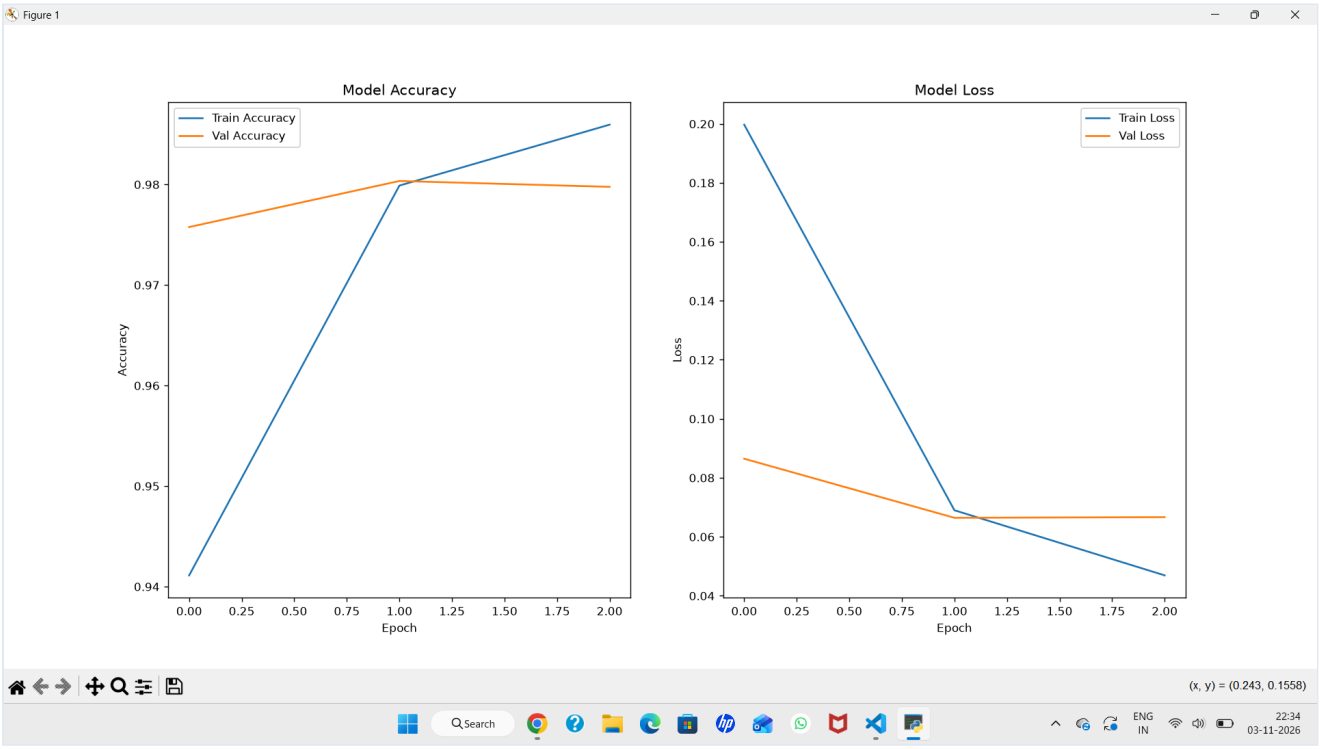


Figure 2: Training and validation accuracy and loss curves demonstrating model convergence.

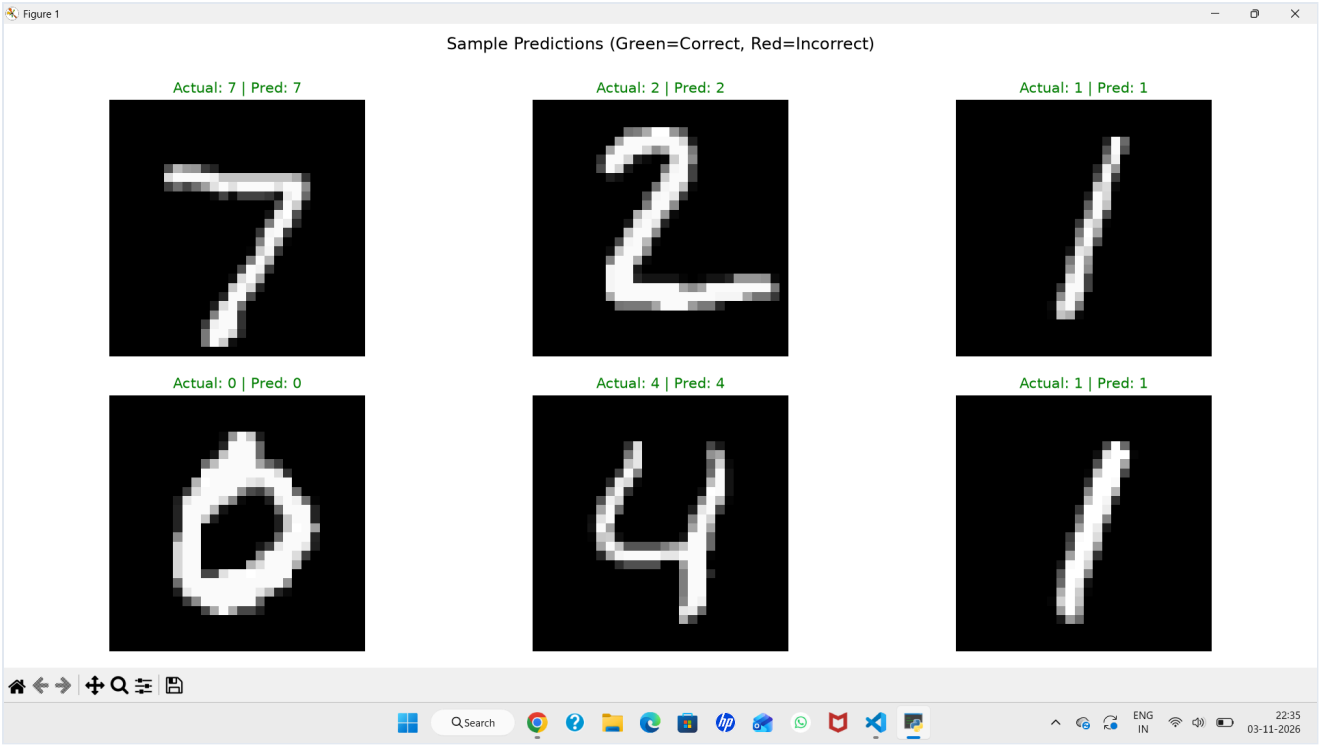


Figure 3: Sample predictions on test data showing accurate classifications of handwritten digits.

3. Streamlit Application (Part B)

An interactive Student Marks Calculator was developed using the Streamlit library. The purpose of this application is to accept numerical grades from the user for three different subjects, calculate the total score out of 300, determine the overall percentage, and output a final Pass or Fail evaluation.

The interface includes basic input validation (restricting inputs to values between 0 and 100) and displays the calculated outputs using clean metric blocks and dynamic success/error alert messages based on the final percentage.

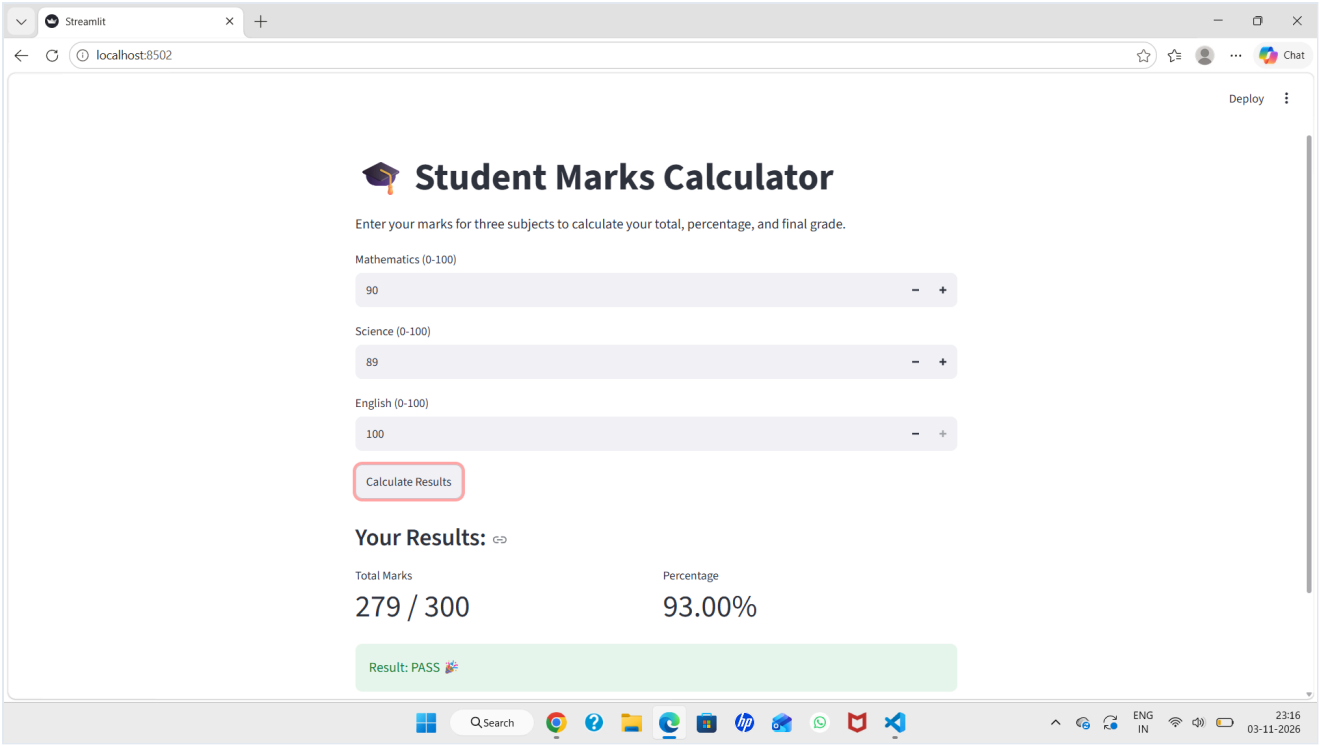


Figure 4: The functioning Student Marks Calculator running locally, displaying accurate input processing and conditional formatting.

4. Results

CNN Performance: The Convolutional Neural Network performed exceptionally well on the MNIST dataset. After only 3 epochs of training, the model achieved a validation accuracy of over 98% and successfully predicted unseen handwritten digits with minimal loss.

Streamlit Performance: The web application successfully integrated with the local Python environment. It renders a clean user interface, actively captures numerical inputs, computes accurate aggregated metrics, and correctly executes the conditional logic to award passing or failing grades without any execution errors.

5. Conclusion

This project successfully demonstrated the end-to-end workflow of building both an artificial intelligence model and a functional software interface. Building the CNN highlighted the importance of data normalization and layer architecture in computer vision tasks. Developing the Streamlit application provided practical experience in deploying Python logic into an accessible format. A potential future improvement would be integrating the trained CNN directly into the Streamlit app, allowing users to draw a digit on the webpage and have the model predict it in real-time.